Fissure Gas Samples, April 2026 – Notes for Phillip

Steamboat Hills Geothermal Outflow Monitoring – UNR Thesis Project

2026-10-15

Background

Steamboat Hills has a fairly long geochemistry record for a Great Basin system. White and coworkers mapped and characterized it geologically and hydrologically starting in the 1950s-60s, and there's a run of gas and water chemistry papers from the 1970s-90s: Nehring (1979, 1980) on reservoir temperature and gas composition, Mariner and Janik (1995) with water and gas analyses for several CPI production wells – well 23-5 among them, which ends up being our main comparison point below – and Arehart, Coolbaugh, and Poulson (2003) arguing for a magmatic heat source from trace elements and gas geochemistry. I've spent most of this project building a database that pulls all of this together with NDEP, NDWR, and our own field data, mostly for the chloride side of the thesis. But it means we actually have a real, if sparse, historical gas-chemistry baseline to check new samples against, which turned out to matter a lot for what follows.

The 2025 eruption and the fissure

In June 2025 the Lower Sinter Terrace had a geyser-like eruption that opened a new fissure – Lindsey et al. (2026) give the fuller account of how the activity built over 2022-2025 before that happened. The well was capped not long after, but the discharge didn't stop; it just moved laterally along the fracture system, and the fissure has kept degassing and discharging since. Cary and I put together a poster on this for GRC (Irvin and Lindsey, 2026), arguing mostly from paired temperature loggers and a Cl mass-balance estimate that this looks more like redistribution within an existing shallow conduit system than a brand-new source turning on. The gas data below is the first real test of that idea from a completely different angle.

Sampling

You, Tobias, and Karim collected three gas samples at and around the fissure on April 29-30, 2026: two points right at the fissure and a third across the road at a separate degassing spot.

Sample	Date	Approx. location	Field temp (C)	Notes
NV26-1	Apr 29	at the fissure	99.5	co-located with a water sample we collected May 1 (see below)
NV26-2	Apr 30	fissure, second point	96	flagged – possible bottle leak noted in the field, also the point where sampling broke through fresh asphalt that the gas was escaping through
NV26-3	Apr 30	across the road	96	second degassing area

NV26-2 is the one I most want your read on. We're not throwing it out, but we're not treating it the same as the other two either, at least not until we hear back on whether the leak and/or the asphalt breakthrough would meaningfully change the result.

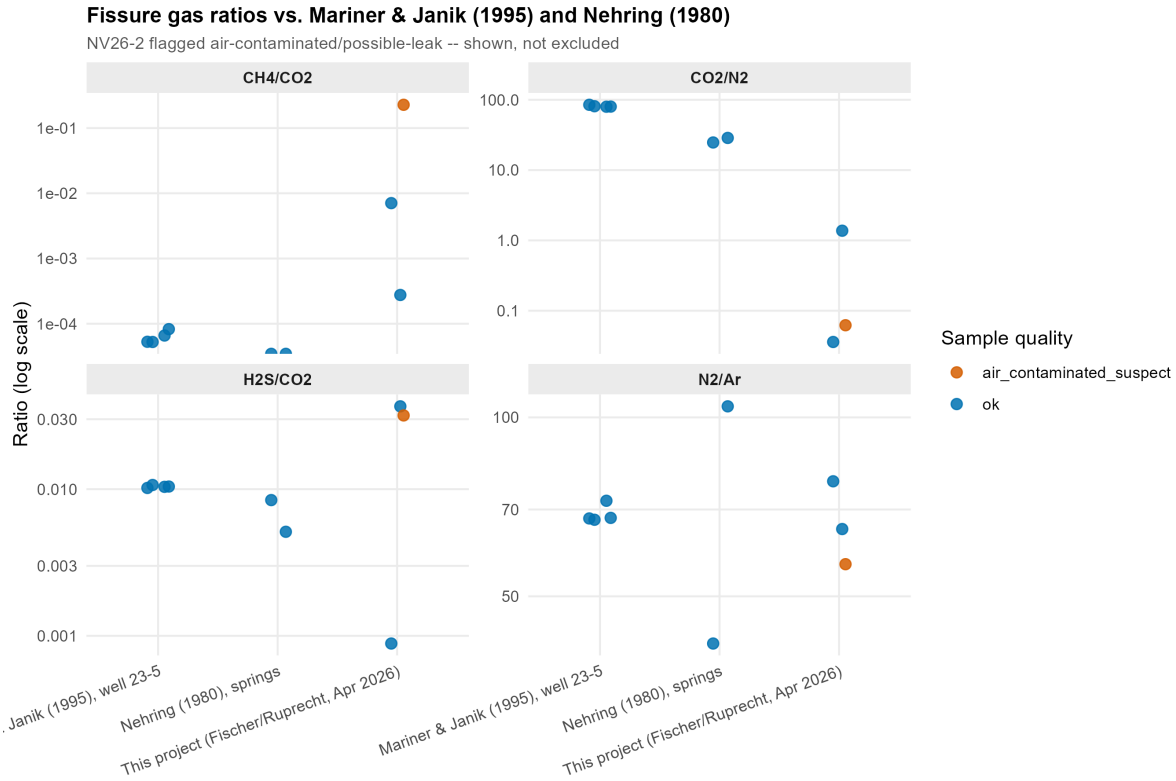
What we're seeing so far

This is exploratory. Three samples, one of them flagged, so everything below is a first look, not a result I'd want to lean on hard yet.

Air correction and geothermometry. Surface gas is never air-free, so before doing anything else we corrected for atmospheric contamination – the standard approach: treat all measured O2 as air, subtract air-derived N2 using the dry-air N2/O2 ratio, renormalize everything else. All three samples have a real amount of air mixed in, which isn't surprising for shallow vent sampling rather than a wellhead, but NV26-2 turns out to be essentially explainable by air alone.

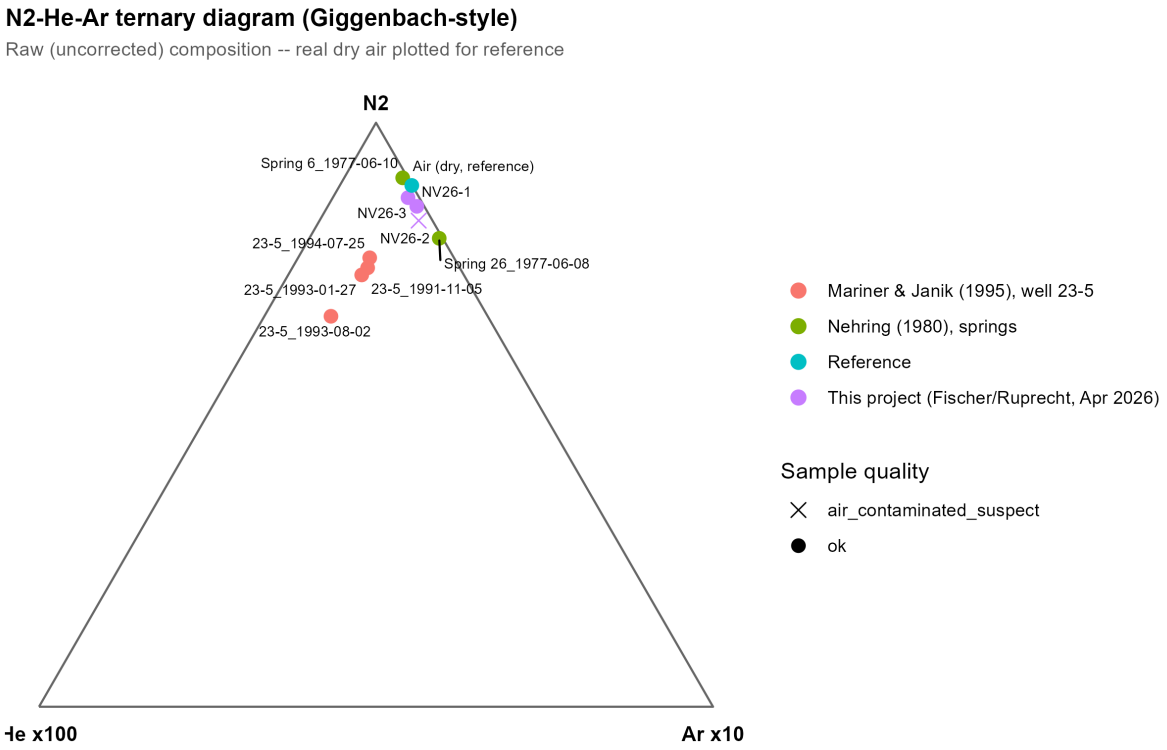
Sample	Air fraction (O2-based)	D'Amore-Panichi T (C)
NV26-1	53%	228
NV26-2	100%	209 (not trusted – see above)
NV26-3	64%	181

Mariner and Janik's well 23-5 T-DP values from the 1990s run 173-200 C, so NV26-1 and NV26-3 bracket that range, and both sit well above the measured field temperatures (96-99.5 C) – the same pattern already showing up in the water geothermometers, where quartz reads below measured temperature and Na/K reads above it. A reasonable result. Three samples still isn't much to build a conclusion on.



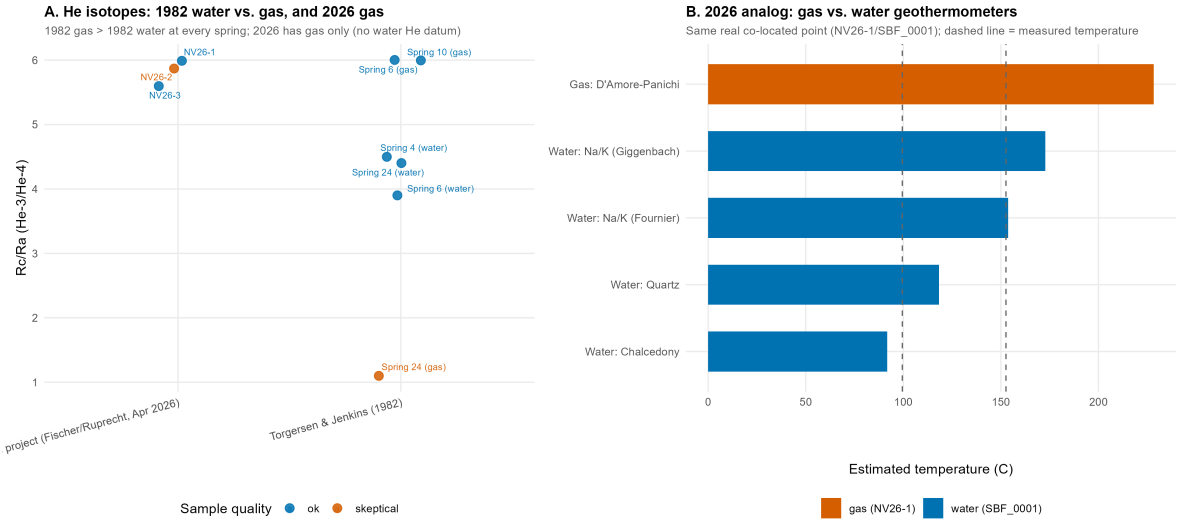
How the two "good" samples compare to Nehring and to Mariner & Janik. CO2/N2, H2S/CO2, N2/Ar, and CH4/CO2 for NV26-1 and NV26-3 land closer to Nehring's (1980) spring samples than to well 23-5, which makes sense – Nehring sampled surface springs too, not a wellhead, so a similar degree of air mixing isn't a surprise. NV26-2 sits off on its own in every panel, again consistent with it being mostly air. Nehring and Mariner & Janik only overlap with us on major-gas composition; neither reports a helium isotope ratio, so that comparison stops at CO2/H2S/N2/Ar/CH4.

A ternary view of the same thing. The classic way to look at an air-mixing question like this is an N2-He-Ar diagram – the same style Mariner & Janik use in their own Figure 9. I plotted our three points against well 23-5, the two usable Nehring rows, and real dry air for reference.



All three of our points, and both of Nehring's spring samples, cluster right next to the air point. Well 23-5 sits noticeably further away, toward more N2 relative to Ar and He, which lines up with the air correction above – surface vent and spring gas looks a lot more air-like than gas collected at a wellhead. Not a new idea, but it's nice to see it fall out of the data this cleanly. I used the common He-times-100/Ar-times-10 scaling to make the plot readable; I haven't gone back to a primary Giggenbach reference to confirm that's the right scaling, or where the "real" air/crustal/mantle field boundaries are supposed to sit on a plot like this, so that's flagged as a question below rather than presented as settled.

Helium isotopes – there's a real 1982 baseline, and it's a surprisingly tight match. I'd first written this section saying nobody had ever published a He-3/He-4 ratio for Steamboat. Wrong – and I'm glad whoever caught that did, because the real source turned out to be better than what I'd patched it with the first time. Torgersen and Jenkins (1982, GCA) measured He isotopes in both water and gas at four Steamboat springs. The two gas samples they themselves call more reliable (Springs 6 and 10) both come out at Rc/Ra = 6.0. Our three new samples come out at 5.6-6.0. That's about as close a match as you could ask for across 44 years, and it's a gas-to-gas comparison specifically, which matters, since their own water-phase numbers run lower (3.9-4.5) and two of their gas values are flagged in their own text as suspect (air mixing, or the vent contaminating itself). See the figure below. Kennedy and van Soest (2007, Science) – which is actually where I first found this, via Dhakal et al.'s citation of it, before going back and reading the primary paper myself – also names Steamboat as a high-R/Ra Basin and Range site, but that's a regional paper, not Steamboat-specific. Torgersen and Jenkins is the real, direct comparison. Arehart et al. (2003) make basically the same magmatic-source argument from trace elements and gas geochemistry, but cite a different, earlier Kennedy paper (2002, not the 2007 one) and don't give a gas composition table or a He number of their own, so they're interpretive context here rather than a number to compare against.



Panel A above is the He-isotope comparison. Panel B is the closest thing I can show for 2026, since we don't have a dissolved-gas/He measurement on the water side this time – it's the gas vs. water temperature geothermometer comparison at this same NV26-1/SBF_0001 point instead. The pattern rhymes even though the measurement is different: in 1982 every spring's gas phase reads a higher Rc/Ra than its own water phase, and in 2026 the gas geothermometer reads hotter than every water method except Na/K. Different quantity, same qualitative story – the gas phase is consistently telling you something deeper and hotter than the water at the same point, in both eras.

NV26-1 sits right on top of SBF_0001, a water sample we collected two days later (May 1) with full major-ion and isotope chemistry – a mature Na-Cl thermal water, Cl 815 mg/L. It's the first spot in the project with both phases from essentially the same location, which matters more for the mixing modeling below than for anything already covered above.

How NV26-1, NV26-2, and NV26-3 relate to each other underground

Table 1

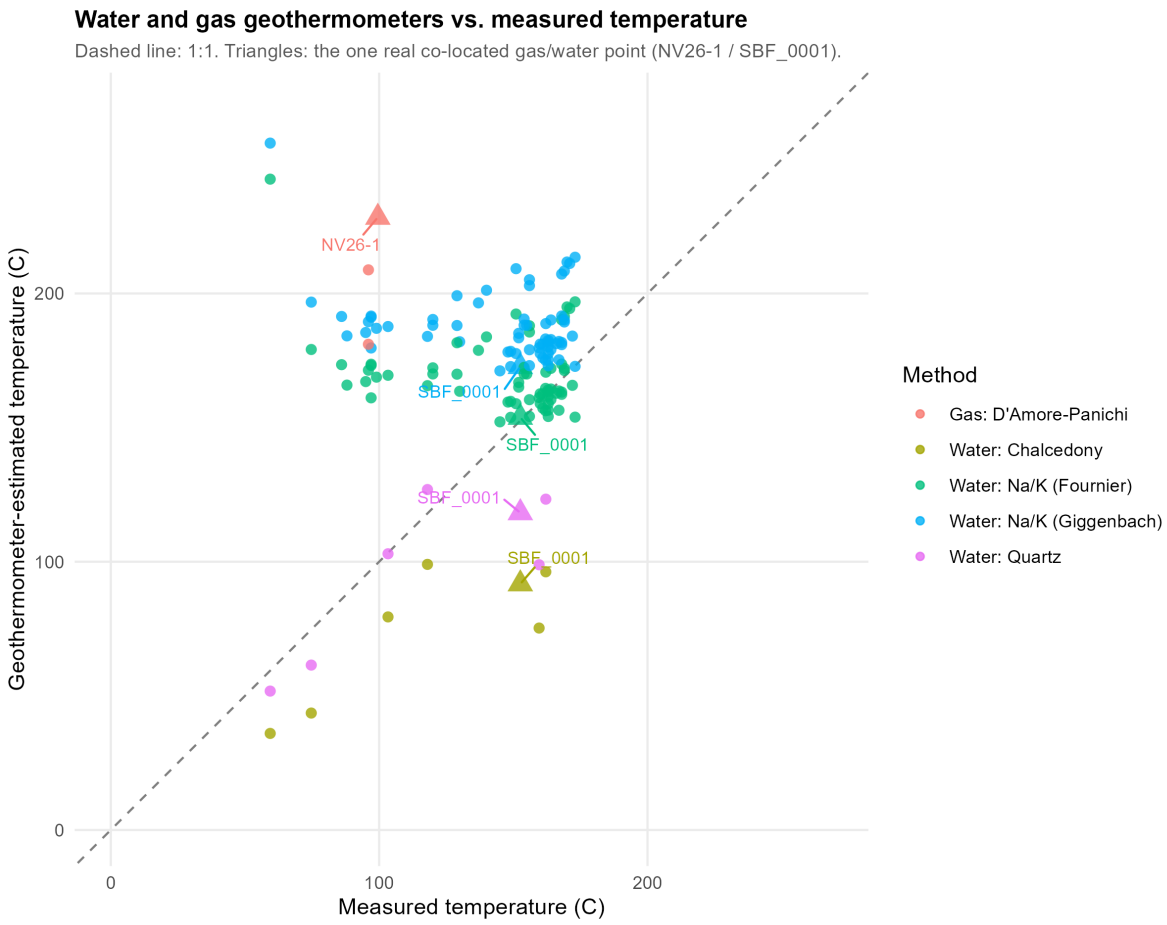
Sample	Field T (C)	CO2, air-corrected (%)	H2S/CO2, air-corrected	He R/Ra	d13C (permil)	D'Amore-Panichi T (C)
NV26-1	99.5	95.2	0.037	5.989	-8.28	228
NV26-2	96.0	54.8	0.032	5.873	-9.49	209
NV26-3	96.0	8.1	0.001	5.600	-9.16	181

NV26-1 and NV26-2 both come from the fissure itself; NV26-3 is across the road, a separate degassing spot. Setting NV26-2 aside for a moment, since it's almost entirely air by the correction above and therefore weak evidence either way, the real trend running from NV26-1 to NV26-3 is a decline in both He R/Ra (5.989 to 5.600) and D'Amore-Panichi temperature (228 C to 181 C), alongside a shift in carbon isotope composition (d13C -8.28 to -9.161 permil, getting slightly more negative).

I can read that two ways and I genuinely don't know which is right. One reading is a single reservoir feeding both points, with progressive cooling and/or dilution along whatever shared path the gas takes to reach each vent – the fissure vent being closer to the "source" in some sense, the across-the-road vent being a bit further along and a bit more diluted. The other reading is that these are two separate vents tapping a similar but genuinely distinct source at depth, and the small differences we're seeing are just that: two different gas parcels that happen to look alike rather than one parcel that evolved along the way. Three samples across two sites isn't enough to tell those apart on its own, at least not with anything I know how to do. Do you have an intuition, from other systems you've worked on, for which of those two is more typical when you see a gradient like this between two nearby vents – and is there a diagnostic (a ratio, an isotope system, something in the noble-gas data we already have) that would actually distinguish them if we had more samples?

Water vs. gas geothermometry, and what the stacking actually means. Putting the gas geothermometer estimate next to our existing water geothermometers (Na/K, quartz, chalcedony – 148 estimates across 67 real thermal samples) for the one point where we have

both, NV26-1's gas estimate (228 C) lands closest to the Na/K water estimates (153-173 C), not the quartz/chalcedony ones (92-118 C).



We've shown this stacking before without really saying what it means, so here's our actual read on it. Gas geothermometer hotter than Na/K, Na/K hotter than quartz and chalcedony – that ordering is the classic signature of boiling and phase separation. When a thermal fluid boils on its way up, gas partitions preferentially into the vapor phase, and that vapor can carry a hotter, deeper equilibrium signature with it even as the residual liquid keeps re-equilibrating (and diluting) on the rest of the trip to the surface. That would also explain quartz sitting below measured temperature – silica re-equilibrates fast and is tracking a shallower, cooler stage of the same ascent than Na/K is. It's a tidy story and it fits, but it's our working interpretation, not something we've tested against an alternative. Is that the right read, or is there a simpler explanation we're missing – for instance, a known issue with running

D'Amore-Panichi on samples that needed this much air correction in the first place?

What kind of fumarole is this?

Worth asking plainly, since it bears on everything above: is this a magmatic vent or something shallower? The gas is CO2-dominated with only a few percent H2S, the field temperatures sit right around local boiling (96-99.5 C, not superheated), and nothing in the geologic literature we've got points to a shallow magma body under Steamboat. Taken together, that reads to us as a peripheral, steam-heated fumarole – boiling of ascending thermal water near the surface, rather than gas coming off a magmatic source directly. That's a fairly ordinary call for a system like this, but it's also exactly the kind of thing we'd rather have you confirm than assume. Does that classification hold up, or is there something in the composition here that argues against it?

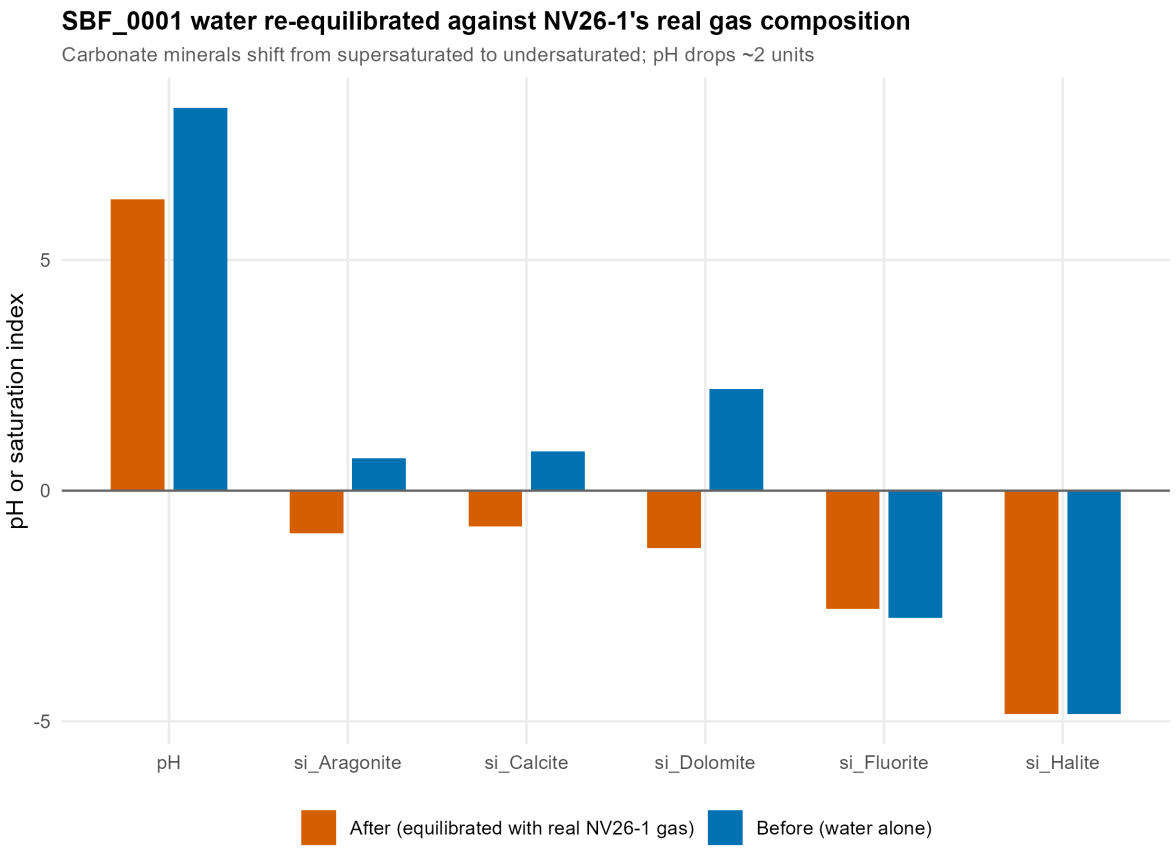
Deep source, shallow expression? Pulling it together

Stepping back, here's how we're currently reading the whole picture, for what it's worth:

- Na/K (153-173 C) and the gas geothermometer (181-228 C) both point to a genuinely deep, hot reservoir underneath this system.
- Quartz and chalcedony (92-118 C), together with the measured vent temperature sitting right at local boiling, point to substantial cooling and/or dilution happening close to the surface.
- The He R/Ra (5.6-6.0) matches Torgersen and Jenkins' trusted 1982 gas values (6.0) about as closely as two measurements 44 years apart could match, which reads to us as a deep, long-lived, mantle-influenced source that hasn't meaningfully changed in that time.

Put together, that's "deep source, shallow boiling expression" – a hot, isotopically stable reservoir at depth, expressed at the surface through a near-boiling, actively cooling and diluting flow path. That's our synthesis from what we have so far, not a finding we'd defend hard on three gas samples. Does that match how you'd read it, or is there a piece of this that doesn't sit right to you?

We also ran a real PHREEQC gas-phase equilibration on this pairing, described in detail below, and it complicates this picture a little rather than confirming it cleanly – worth reading before you weigh in on the synthesis above.



The PHREEQC run, in more detail. We went ahead and ran a real PHREEQC `GAS_PHASE` equilibration rather than just talking about it in the abstract. The setup: take SBF_0001's full measured water chemistry (the May 1 water sample, co-located with NV26-1), and re-equilibrate it against a gas phase seeded with NV26-1's real, air-corrected CO2:H2S ratio – about 96.5:3.5 – at 1 atmosphere. We treated it as an open system, meaning the gas phase is allowed to keep resupplying itself rather than being a small, fixed pocket that gets used up, on the assumption that the fissure is more like a continuous vent than a sealed reservoir.

The result: pH drops from the water's measured 8.3 down to a modeled 6.3, and calcite, aragonite, and dolomite all flip from supersaturated to undersaturated once the gas is added. That's a real, fairly large shift, and it means the water as sampled isn't sitting in equilibrium with the gas as sampled under the assumptions we used. The honest question is what that actually tells us. It could mean the water and gas genuinely aren't the same fluid parcel – that gas and liquid separate out at depth or on the way up, before either one reaches the surface, which would actually support the boiling story above rather than undercut it. Or it could mean our modeling choices are driving the result more than the real chemistry is: 1 atm was a simplification (we don't have a real estimate of the pressure this gas phase should represent

at depth), and the open-system assumption could be wrong in either direction. I don't have the geochemical modeling background to know which of those is doing more of the work here, and that's really the core question I want your take on – is a disequilibrium result like this expected and unremarkable for a water and gas sampled two days and some distance apart, or is it telling us something real about how this system separates gas from liquid?

How this fits into the rest of the thesis

This is the first real gas dataset the project has had, and it slots into a few places we'd already scoped out but had no data for:

- The D'Amore-Panichi geothermometer code had only ever been run against your group's own historical numbers (Mariner & Janik) as a check – this is the first time it's run on a Steamboat sample we actually collected ourselves.
- We'd had a PHREEQC gas-phase/mixing setup built for a while, waiting on a real dissolved-gas sample to test it against. NV26-1 is the first one, and the run above is already a real result, not just a working capability anymore.
- The NV26-1/SBF_0001 pairing is a natural input to the mixing-model chapter: a real gas end-member next to a real water end-member from (almost) the same spot, the same week.
- More broadly, it's a second, independent way to ask whether the system has cooled and diluted the way the water geochemistry already suggests – which matters directly for the redistribution-vs-new-source question the poster raised.

Questions for you

Mostly I want to make sure we're using and describing your group's work correctly before any of this goes further, but a few of these are genuinely open scientific questions where I'd value your read more than anything else:

- How should we cite the gas analyses in the thesis, and separately if this turns into a conference abstract or paper – Fischer's lab and Barry's lab specifically, or is there a preferred citation you'd want us to use?
- Is the air correction above a reasonable way to handle Giggenbach bottle samples collected this way, or does your group normally handle it differently, and should we match that instead?
- On the PHREEQC gas-phase run: is 1 atm/open-system a reasonable starting assumption, or is there a better estimate of the real pressure and volume this gas phase should represent at depth? And is the disequilibrium we got (pH and carbonate SI both shifting a lot) actually surprising, or is that about what you'd expect for a water and gas sampled two days and some distance apart rather than from the exact same parcel?

- On NV26-1 vs. NV26-3 (and setting NV26-2 aside): does the decline in He R/Ra and gas temperature between the two look to you like a single reservoir feeding both points with cooling/dilution along the way, or two separate vents on a similar-but-distinct source? Is there a ratio or isotope system in the data we already have that would actually distinguish those two possibilities?
- On the gas-over-water geothermometer stacking: does the boiling/phase-separation read we laid out above sound right to you, or is there a simpler explanation – a known issue running D'Amore-Panichi on heavily air-corrected samples, for instance?
- Does "peripheral, steam-heated fumarole" sound like the right call for this vent, given the CO2-dominant, low-H2S composition and the near-boiling field temperatures, or is there something here that points toward (or against) a more direct magmatic connection?
- Does "deep, hot, isotopically stable source; shallow, boiling, diluting expression" match how you'd read this dataset, or do you see it differently?
- On the ternary plot: is the He-times-100/Ar-times-10 scaling I used the right convention, and is there a reference version of that diagram, with real air/crustal/mantle field boundaries, that we should be plotting these points against instead of just showing them relative to a single air point?
- On NV26-2: is the leak serious enough that the sample shouldn't be used at all, or is it still usable with the caveat we've got on it?
- Should we hold off on presenting any interpretation of the noble gas and carbon isotope numbers until you, Tobias, or Karim have had a chance to look at them yourselves, or is what's here reasonable to go forward with as a preliminary read?
- Would you, Tobias, and/or Karim want to be coauthors if this turns into a conference abstract or paper down the line?

Contact

Tyler Irvin, UNR – happy to send the underlying data and derived CSVs if useful on your end.